

Binomial Models For Option Pricing

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Options

- ▶ With Forwards and Futures, both parties are **obligated** to transact at maturity T .
- ▶ What if you only want to transact if the market moves in your favor?

What is an Option?

A contract that gives the buyer the **right, but not the obligation**, to buy or sell an underlying asset at a specified strike price (K) on or before a specified date (T).

Call Option

- ▶ The right to **BUY**.
- ▶ You want the asset price (S_T) to go **UP**.

Put Option

- ▶ The right to **SELL**.
- ▶ You want the asset price (S_T) to go **DOWN**.

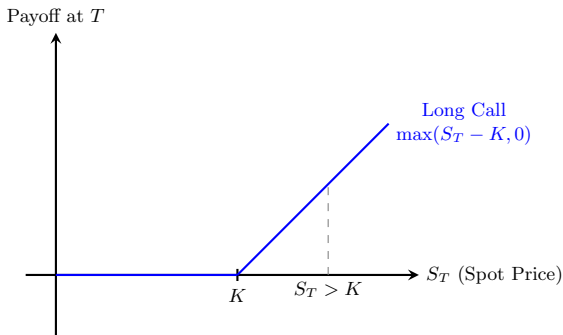
The Catch: The Premium

Because the buyer has a choice (and the seller has all the risk), entering an option contract is **not free**. The buyer must pay a price upfront at $t = 0$, called the **Option Premium**. Our goal is to figure out how to calculate this premium!

Call Option: Payoff Profile

A **Call Option** gives the holder the *right, but not the obligation*, to buy the underlying asset at a specified strike price (K) at maturity (T).

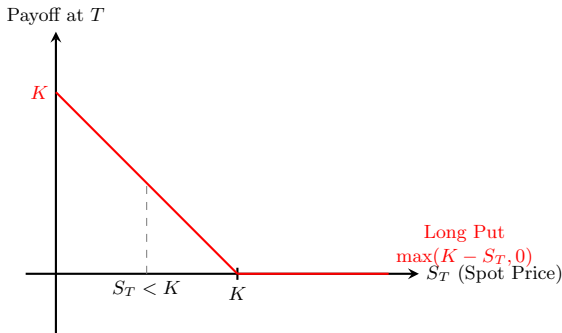
- ▶ **Long Call Payoff:** $\max(S_T - K, 0)$
- ▶ **Logic:** If $S_T > K$, the option is exercised (buy at K , worth S_T). If $S_T \leq K$, the option expires worthless (loss is limited to the initial premium paid, which is not shown in the raw terminal payoff).



Put Option: Payoff Profile

A **Put Option** gives the holder the *right, but not the obligation*, to sell the underlying asset at a specified strike price (K) at maturity (T).

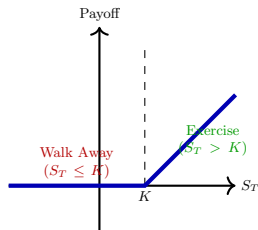
- **Long Put Payoff:** $\max(K - S_T, 0)$
- **Logic:** If $S_T < K$, the option is exercised (sell at K , even though market is lower). If $S_T \geq K$, the option expires worthless.



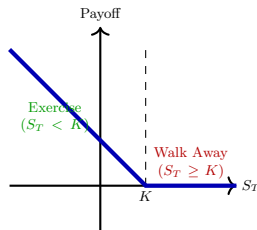
Option Payoffs: Non-Linear Structure

- At maturity T , the option buyer will only exercise the contract if it is profitable. Otherwise, they walk away (payoff is zero).

$$\text{Long Call Payoff} = \max(S_T - K, 0)$$



$$\text{Long Put Payoff} = \max(K - S_T, 0)$$

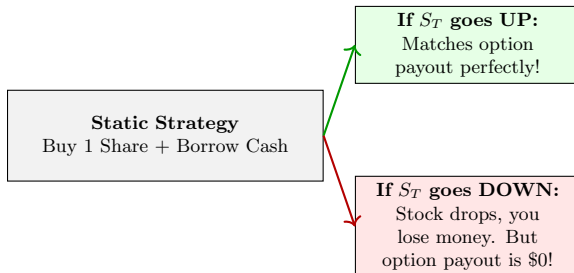


The Asymmetry

Unlike Forwards, the payoff is **kinked** at the strike price K . The downside is strictly limited to zero, while the upside can be massive.

The Breakdown of Static Replication

- **Recall:** We priced Forwards easily using a “Cash and Carry” strategy (borrow money, buy 1 share of stock today, hold until T).
- **Can we do this for a Call Option?**



The Pricing Dilemma

You **cannot** statically mix a straight line (stock) and a flat line (bond) to perfectly match a kinked curve (option) across all possible futures. The static arbitrage argument fails!

A Call Option's payoff is conditional: $\max(S_T - K, 0)$. If we had a magic oracle at $t = 0$ telling us the future, static replication would be easy.

World A: Oracle says it finishes ITM

- **Condition:** $S_T > K$
- **Payoff:** Exactly $S_T - K$
- **Static Replication:** Same as a Forward contract! Buy 1 unit of asset, borrow Ke^{-rT} .
- **Cost today:** $S_0 - Ke^{-rT} > 0$

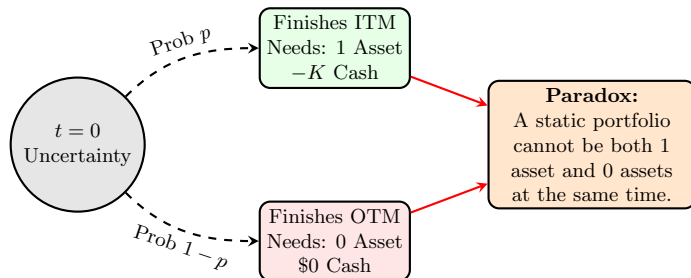
World B: Oracle says it finishes OTM

- **Condition:** $S_T \leq K$
- **Payoff:** Exactly 0
- **Static Replication:** Do nothing. Hold 0 units of the asset, borrow \$0.
- **Cost today:** \$0

The Catch: In these hypothetical worlds, the payoff is linear ($S_T - K$ or 0). Linear payoffs are perfectly replicated statically.

The Reality: Why Static Replication Fails

In reality, no oracle exists. At $t = 0$, we face **uncertainty** and must prepare for both possibilities simultaneously.



- ▶ If you build the ITM portfolio statically, and the market crashes, you lose S_0 (Option buyer only loses premium).
- ▶ If you build the OTM portfolio statically (do nothing), and the market skyrockets, you are exposed to infinite risk.

One Period Binomial Model

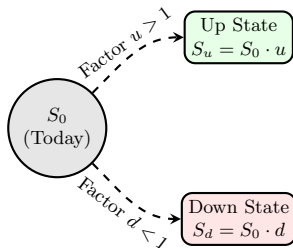
Introduction to the Binomial Model

Binomial Pricing Model (Cox-Ross-Rubinstein, 1979) provides a discrete-time framework

- Price options
- Understand hedging

The Core Assumption: A stock today trading at S_0 can be in two states in one time period

- It goes **Up** by a factor u .
- It goes **Down** by a factor d .



The One-Period Setup (Parameters)

Define an economic environment where no trivial arbitrage exists

- ▶ S_0 : Initial stock price.
- ▶ r : The risk-free interest rate over the period. \$1 invested today grows to $\$(1 + r)$ at $t = 1$.
- ▶ u and d : The up and down multipliers.

The Fundamental No-Arbitrage Condition: For the model to be valid, the risk-free return must lie strictly between the down factor and the up factor:

$$d < 1 + r < u$$

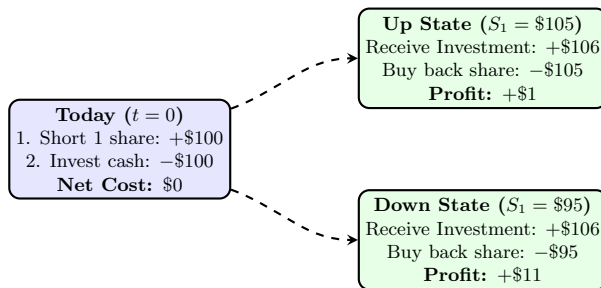
Why?

- ▶ If $1 + r \geq u$: The risk-free bank always beats the stock, even if the stock goes up. (Short stock, invest cash).
- ▶ If $1 + r \leq d$: The stock always beats the bank, even if the stock crashes. (Borrow cash, buy stock).

The Arbitrage Scenario : $1 + r \geq u$

Example Setup: $S_0 = \$100$, $u = 1.05$ (Up to \$105), $d = 0.95$ (Down to \$95)

Interest Rate: $r = 6\% \implies 1 + r = 1.06$ (**Note:** $1.06 > 1.05$)

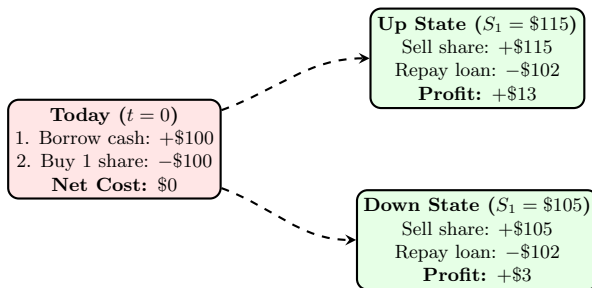


Result: A guaranteed, risk-free profit with zero initial capital. Market forces will instantly eliminate this "money pump." Thus, $1 + r < u$ must hold.

The Reverse Arbitrage : $1 + r \leq d$

Example Setup: $S_0 = \$100$, $u = 1.15$ (Up to \$115), $d = 1.05$ (Down to \$105)

Interest Rate: $r = 2\% \implies 1 + r = 1.02$ (**Note:** $1.02 < 1.05$)

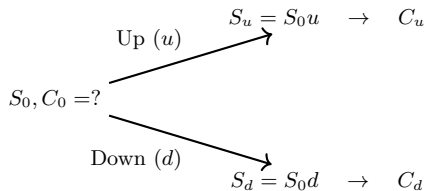


Result: A guaranteed, risk-free profit with zero initial capital. Market participants will borrow massively, driving rates up and stock prices down until $d < 1 + r$ is restored.

Payoff in One-Period Binomial Framework

Assume a stock today trading at S_0 .

$$\begin{cases} S_u = S_0 \cdot u & \text{with payoff } C_u = \max(S_u - K, 0) \\ S_d = S_0 \cdot d & \text{with payoff } C_d = \max(S_d - K, 0) \end{cases}$$



The Replicating Portfolio

We want to price a European Call option (C_0) with strike K . At $t = 1$, its payoffs are known:

- ▶ Up State: $C_u = \max(S_u - K, 0)$
- ▶ Down State: $C_d = \max(S_d - K, 0)$

We construct a **Replicating Portfolio** today ($t = 0$) using the two basic assets:

1. Buy Δ **shares** of the stock.
2. Invest/Borrow B **dollars** at the risk-free rate r .

The total cost today is: $C_0 = \Delta S_0 + B$

We force this portfolio to exactly match the option's payoff at $t = 1$ in *both* states:

State at $t = 1$	Portfolio Value	Option Payoff
Up (S_u)	$\Delta \cdot (uS_0) + B(1 + r)$	$= C_u$
Down (S_d)	$\Delta \cdot (dS_0) + B(1 + r)$	$= C_d$

We now have a system of 2 equations with 2 unknowns (Δ and B).

Solving for Delta (Δ) and Bond (B)

To eliminate all risk, match the portfolio payoff to the option payoff in **both** states:

$$\begin{cases} \Delta S_u + B(1+r) = C_u & \text{(Up State)} \\ \Delta S_d + B(1+r) = C_d & \text{(Down State)} \end{cases}$$

Subtracting the second equation from the first yields the fundamental option hedge ratio:

$$\Delta = \frac{C_u - C_d}{S_u - S_d} = \frac{C_u - C_d}{S_0(u - d)}$$

Interpretation of Delta:

- ▶ It is the ratio of the spread in option values to the spread in stock prices.
- ▶ It represents the **sensitivity** of the option to the stock price.
- ▶ It is exactly the number of shares you must hold today to statically hedge the option over this single period.

Solving for the Cash Position (B)

We substitute $\Delta = \frac{C_u - C_d}{S_0(u - d)}$ back into the Up-State Equation (Eq. 1):

$$\Delta \cdot (uS_0) + B(1 + r) = C_u$$

$$\left[\frac{C_u - C_d}{S_0(u - d)} \right] uS_0 + B(1 + r) = C_u$$

$$\frac{u(C_u - C_d)}{u - d} + B(1 + r) = C_u \implies B(1 + r) = C_u - \frac{uC_u - uC_d}{u - d}$$

Simplifying and discounting back to today gives us the exact formula for B :

$$B = \frac{1}{1 + r} \left(\frac{uC_d - dC_u}{u - d} \right)$$

Deriving the Option Price (C_0)

We have

$$C_0 = \Delta S_0 + B$$

Substituting our solved expressions for Δ and B :

$$C_0 = \left[\frac{C_u - C_d}{S_0(u - d)} \right] S_0 + \frac{1}{1 + r} \left[\frac{uC_d - dC_u}{u - d} \right]$$

Grouping the terms by the future option payoffs C_u and C_d :

$$C_0 = \frac{1}{1 + r} \left[\left(\frac{(1 + r) - d}{u - d} \right) C_u + \left(\frac{u - (1 + r)}{u - d} \right) C_d \right]$$

Letting $p = \frac{(1+r)-d}{u-d}$, we get our final pricing formula:

$$C_0 = \frac{1}{1 + r} [pC_u + (1 - p)C_d]$$

The Risk-Neutral Pricing Formula

$$C_0 = \frac{1}{1+r} [pC_u + (1-p)C_d]$$

where $p = \frac{(1+r)-d}{u-d}$

Because $d < 1+r < u$, we know $0 < p < 1$. Therefore, we interpret p as a probability.

- **Why "Risk-Neutral"?** If we assume p is the true probability of an up-move, the expected return of the risky stock exactly equals the risk-free rate ($E[S_1] = S_0(1+r)$).

$$E[S_1] = p \cdot (uS_0) + (1-p) \cdot (dS_0)$$

$$E[S_1] = S_0 [pu + (1-p)d]$$

Substituting $p = \frac{(1+r)-d}{u-d}$ and $1-p = \frac{u-(1+r)}{u-d}$ simplifies the term in the brackets to exactly $(1+r)$:

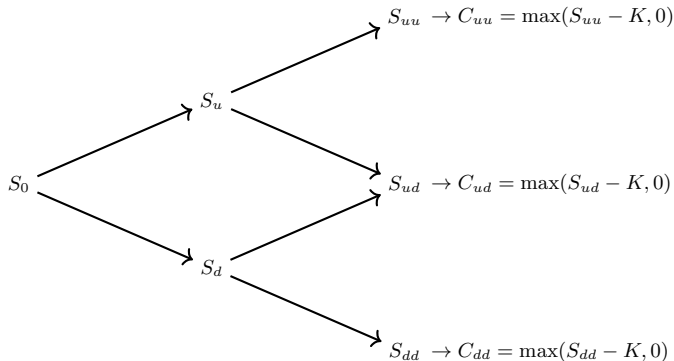
$$E[S_1] = S_0(1+r)$$

- **The Core Insight:** The fair price of the option is simply its *expected payoff* in this hypothetical risk-neutral world, discounted back to today at the risk-free rate.

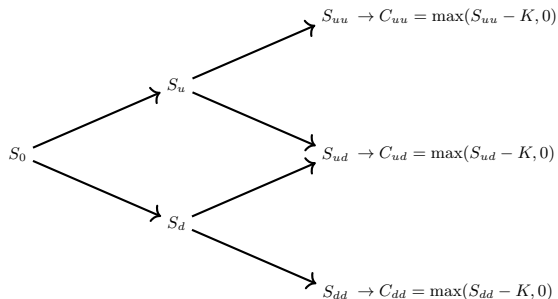
Two Period Binomial Tree

Scaling Up: The 2-Period Binomial Tree

To model realistic time horizons, we concatenate single periods into a multi-step tree ($t_0 \rightarrow t_1 \rightarrow t_2$).



Scaling Up: The 2-Period Binomial Tree



Backward Induction Protocol

1. **Terminal Nodes (t_2):** Evaluate intrinsic option payoffs directly at expiry.
2. **Intermediate Nodes (t_1):** Discount expected risk-neutral values:

$$C_u = \frac{1}{1+r} [pC_{uu} + (1-p)C_{ud}], \quad C_d = \frac{1}{1+r} [pC_{ud} + (1-p)C_{dd}]$$

3. **Root Node (t_0):** Step backward once more to obtain present option price C_0 .

Static replication fails because the required hedge ratio (Δ) changes at every node across the tree!

Hedge Ratio at Node $t = 0$

$$\Delta_0 = \frac{C_u - C_d}{S_u - S_d}$$

Hold Δ_0 shares of stock to stay hedged over $t_0 \rightarrow t_1$.

Hedge Ratio at Node $t = 1$ (Up State)

$$\Delta_u = \frac{C_{uu} - C_{ud}}{S_{uu} - S_{ud}}$$

Because $\Delta_u \neq \Delta_0$, the portfolio is no longer hedged!

Why Option Replication Must Be Dynamic

- ▶ As S_t moves, the likelihood of finishing in-the-money changes $\rightarrow \Delta$ shifts continuously.
- ▶ **Dynamic Rebalancing:** The market maker must actively buy or sell shares at intermediate nodes to maintain a zero-risk portfolio.

Numerical Example: Pricing a Call Option

Let's apply the formulas to a concrete one-period example.

The Setup: Initial Stock = 100, Strike = 105, Interest Rate = 5% ($1 + r = 1.05$)

Up factor = 1.2 (Stock goes to 120), Down factor = 0.8 (Stock goes to 80)

Step 1: Find the Payoffs at Maturity

► Up State Payoff = $\max(120 - 105, 0) = 15$

► Down State Payoff = $\max(80 - 105, 0) = 0$

Step 2: Calculate Risk-Neutral Probability (p)

► $p = ((1 + r) - d)/(u - d)$

► $p = (1.05 - 0.8)/(1.2 - 0.8) = 0.25/0.40 = 0.625$

Step 3: Calculate Price and Delta

► Price = $[p(15) + (1 - p)(0)]/1.05 = [0.625 \times 15]/1.05 = 8.93$

► Delta = $(15 - 0)/(120 - 80) = 15/40 = 0.375$ shares

What next ? ...

On the Choice of u and d ...

The binomial model assumes

$$S_{t+1} = \begin{cases} uS_t, \\ dS_t. \end{cases}$$

Replication and risk-neutral pricing work once u and d are specified. Question:

Why choose

$$u = 1.2, \quad d = 0.8$$

rather than

$$u = 1.5, \quad d = 0.7?$$

Goal

Choose u and d so that the tree reflects the observed volatility of the stock.

The Limit of Discrete Trees: Continuous Convergence

What happens when we shrink the time step $\Delta t \rightarrow 0$ and let the number of steps $N \rightarrow \infty$?

Discrete Binomial ($N = 4$)

Discrete coin flips yield coarse, step-wise payoff probabilities across a small set of outcome nodes.

Continuous Limit ($N \rightarrow \infty$)

By the Central Limit Theorem, the sum of infinitely many tiny binomial steps converges smoothly to a **Lognormal Distribution**.

